

Python Strings & Slicing

A clear beginner-friendly guide with concepts, syntax, examples, and common methods.

1. What is a String?

A string is a sequence of characters enclosed in single quotes (' '), double quotes (" "), or triple quotes ("\" / \"\" \"\"). Strings are used to store text in Python.

```
name = "Python"
message = 'Data Science'
```

2. Creating Strings

Type	Example
Single quotes	name = 'Smrutirekha'
Double quotes	name = "Smrutirekha"
Triple single quotes	text = """Hello\nPython"""
Triple double quotes	text = """"Hello\nPython""""

3. String Indexing

Each character in a string has an index. Python uses zero-based indexing, so the first character has index 0. Negative indexing starts from the end, where -1 is the last character.

```
s = "Python"
```

Character	P	y	t	h	o	n
Positive index	0	1	2	3	4	5
Negative index	-6	-5	-4	-3	-2	-1

```
print(s[0]) # P
print(s[-1]) # n
```

4. What is String Slicing?

String slicing is used to extract a portion of a string. The basic syntax is:

```
string[start : stop : step]
```

start is the starting index, **stop** is the ending boundary (not included), and **step** controls how many positions to move at a time.

5. Basic Slicing Examples

Code	Result	Meaning
s = "Python"		
s[0:2]	"Py"	index 0 up to, but not including, 2
s[2:6]	"thon"	index 2 up to, but not including, 6
s[:4]	"Pyth"	from beginning to index 4
s[2:]	"thon"	from index 2 to the end
s[:]	"Python"	entire string
s[-4:-1]	"tho"	negative-index slice

6. Slicing with Step

```
s = "Python"
s[::2] # "Pto"
s[1::2] # "yhn"
s[::-1] # "nohtyP"
```

A negative step moves from right to left. Therefore, **s[::-1]** reverses the string.

7. Useful String Methods

Method	Example	Purpose
upper()	"python".upper()	Converts to uppercase
lower()	"PYTHON".lower()	Converts to lowercase
strip()	" hello ".strip()	Removes leading/trailing spaces
replace()	"hello".replace("h","H")	Replaces text
split()	"a,b,c".split(",")	Splits a string into a list
join()	"".join(["a","b"])	Joins items into a string
find()	"python".find("t")	Returns the index of a substring
count()	"banana".count("a")	Counts occurrences

8. Important Points to Remember

- Strings are sequences of characters.
- Python string indexes start from 0.
- The last character can be accessed using index -1.

- Slicing follows the pattern start:stop:step.
- The stop index is excluded from a slice.
- Strings are immutable, so individual characters cannot be changed directly.

9. Practice Questions

1. What is a string in Python?
2. What is the index of the first character?
3. What does `s[-1]` return?
4. What is the difference between `s[1:4]` and `s[:4]`?
5. How can you reverse a string using slicing?
6. What does the step value do in slicing?